

Project Title: Pitwall

Team 2

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1. Project Description

1.1 Overview

Pitwall is a web platform that helps grassroots racing teams record, organize, and learn from everything that happens at the track. Professional Formula 1 teams employ hundreds of engineers to turn every session into structured data. Club racing, karting, and junior formula teams (such as F3 & F4) usually have one engineer, a few volunteer mechanics, and 1-2 drivers. Currently, these teams' records live on paper setup sheets, group chats, memory, and basic Excel sheets. Pitwall gives these teams the same kind of organized institutional memory that top teams have, at a fraction of the cost and complexity.

1.2 Motivation

Small teams lose performance and money in three recurring ways. First, they lose setup knowledge: a team finds a fast setup at a track but cannot recall exactly what they ran, or which change made the car faster, when they return a year later. Second, they waste tires. Racing tires lose grip each time they go through a heat cycle (heating up during a session and cooling afterward). Without tracking, teams either run tires past their useful life, which is slower, or discard them too early, which is expensive since a set can cost several hundred dollars. Third, driver feedback gets disconnected from results. A driver reports that the car "pushed in turn 3," the engineer changes something, and nobody links the complaint, the change, and the resulting lap times together.

1.3 Primary Goal

Pitwall's primary goal is to make sure that every track session produces reusable, reliable knowledge and data. The system will store each session's car, driver, setup, tires, lap times, and debrief as connected records, so a team can answer questions such as "What was our fastest setup on X track? What is the most optimal setup when it's raining?" or "How many useful heat cycles does this tire set have left?" in a few clicks.

1.4 Innovative Idea

The primary goal of Pitwall is to make sure every track session produces reusable, reliable data. The system stores each session's car, driver, setup, tires, lap times, and debrief as connected records, so a team can answer questions such as "What was our fastest setup at Laguna Seca?", "What setup worked best in the rain?", or "How many useful heat cycles does this tire set have left?" in a few clicks.

1.5 Major Stakeholders

Drivers and race team engineers/managers who want to test specific setups and tire configurations. Team admins that control budget, rosters, equipment.

1.6 Application Domains

Motorsport operations and engineering data management, specifically grassroots and junior categories: karting, club racing (e.g., SCCA, NASA), spec series, and junior open-wheel formulas such as F4 and F3. From a business point of view, this market is much larger than the ten Formula 1 teams: there are thousands of small teams worldwide that currently have no affordable data platform.

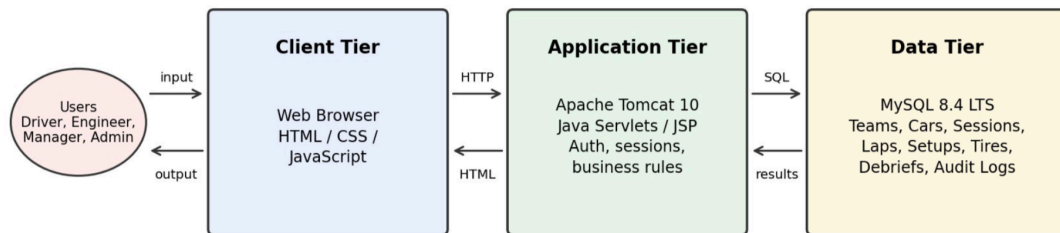
1.7 Benefits to users

Teams save money by using tires to their full useful life, arrive at each track with previous baseline setups, run more productive meetings due to specific setup feedback, and keep their knowledge even when engineers, mechanics, and drivers join and leave the team.

2. System Environment

2.1 System Structure

Pitwall will be structured as a 3-tier client-server architecture



Client Tier - Drivers, engineers, managers, & administrators will be able to access Pitwall through a web browser on a laptop, tablet, or phone. This application will be produced with HTML, CSS, & JS for an interactional frontend.

Application Tier - The application logic will run on Apache Tomcat using Java Servlets and JSP. This tier in the architecture will handle admin login, session management, lap statistics, and setup comparisons.

Data Tier - MySQL will store all data related to cars (tire temperature, engine temperature, etc), car setup (front/rear wing angles, side-pod configuration, etc), sessions, debriefs, and rosters

2.2 Hardware & Software

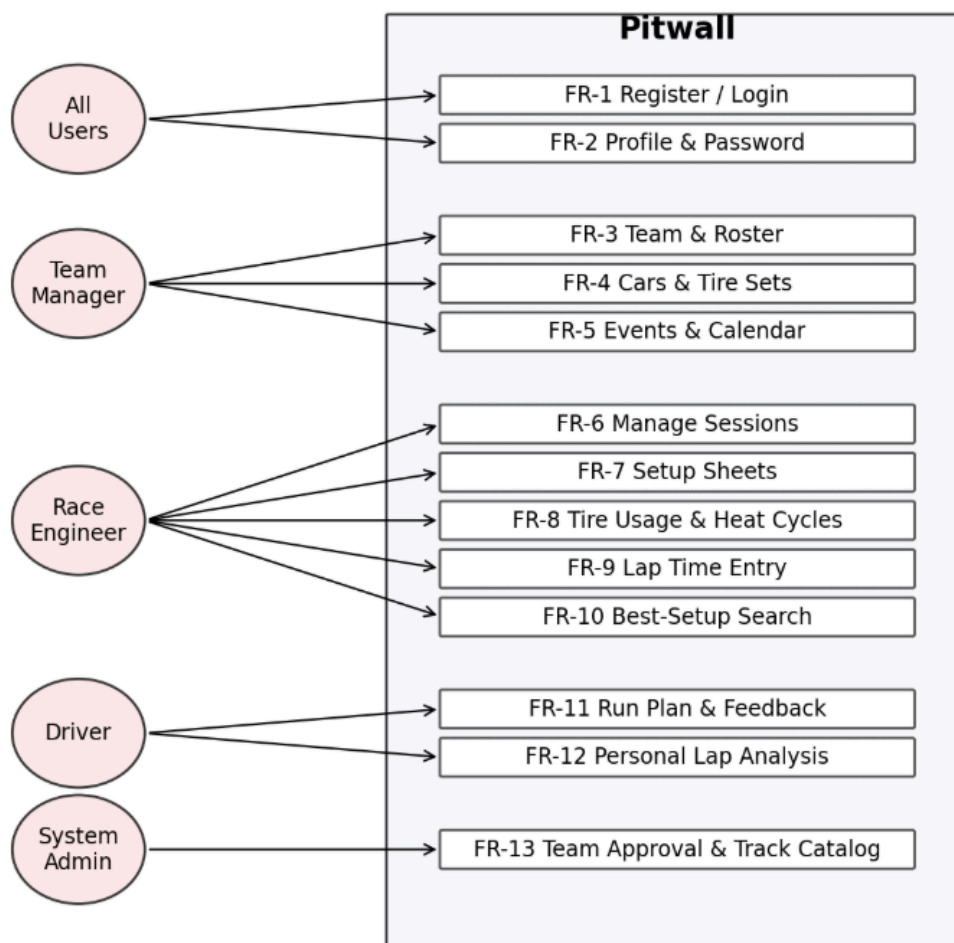
Component	Choice
Server OS	Team development on Windows 11 / macOS
Web / App Server	Apache Tomcat 11
RDBMS	MySQL Community Server V26.7

Application Languages	Java, XML, SQL
Hardware (min.)	Server: 2 vCPU, 4 GB RAM, 40 GB SSD. Client: any device with a modern browser (Chrome, Firefox, Safari, etc...)
Tools	MySQL Workbench, Eclipse IDE, Git + GitHub

3. Functional Requirements

3.1 Users and How They Interact with Pitwall

Every user belongs to one team (except the system administrator) and has one role inside that team. A team is created by a team manager and approved by the system administrator. Figure 2 shows which functions each role uses.



- **Driver:** Views the run plan for each session, submits feedback after the session, and reviews personal lap times. Drivers cannot change setups or team data.
 - **Race Engineer:** Creates sessions, records setup sheets, records tire set usage, enters lap times, and searches past setups.
 - **Team Manager:** Manages the team roster, cars, tire sets, and event calendar. A manager also has all engineer permissions.
 - **System Administrator:** Approves or suspends team accounts and maintains the shared track catalog. The administrator does not view teams' private engineering data.
- The database will manage 10 tables: Team, User, Car, Track, Event, Session, SetupSheet, TireSet, SessionTireUse, and Lap.

3.2 All Users

FR: 1 Account Registration and Login

Description	Users create an account and log in before using any function.
Input	Name, email, password; a team invite code (for drivers/engineers) or a new team name (for managers).
Process	Validate input, check the email is unique, hash the password, INSERT into User (and INSERT a pending Team for a new manager); on login, SELECT the user, verify the hash, and create a server session holding user ID, team ID, and role.
Output	New account or logged-in dashboard for the user's role; error message on failure.

FR: 2 Profile and Password Management

Description	Users view and update their own profile.
Input	Display name, phone, driver number (drivers), current and new password.
Process	Verify the current password, validate fields, UPDATE the User row.
Output	Updated profile and confirmation message.

3.3 Team Manager

FR: 3 Team and Roster Management

Description	The manager builds and maintains the team.
Input	Team name, series/category; member email and role (Driver, Engineer, Manager).
Process	UPDATE Team; generate invite codes; UPDATE member roles in User; deactivate (not delete) members who leave so their history stays linked.
Output	Current roster list with roles and status.

FR: 4 Cars and Tire Sets Management

Description	Record each car and each tire set the team owns.
Input	Car: chassis number, model, class, engine, weight, notes. Tire set: label/barcode, brand, compound, purchase date, price, maximum heat cycles.
Process	INSERT/UPDATE/deactivate Car and TireSet rows linked to the team; new tire sets start with 0 heat cycles and status New.
Output	List of cars (with session count and last-used date) and tire inventory (with status New, In Use, or Retired).

FR: 5 Event and Calendar Management

Description	Plan race weekends and test days.
Input	Event name, track (from catalog), start/end date, event type (test, race), entry fee.
Process	INSERT Event linked to Team and Track; list upcoming and past events ordered by date.
Output	Team calendar; each event links to its sessions.

3.4 Race Engineer

FR: 6 Session Management

Description	Create a record for each on-track session.
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Input	Event, session type (Practice, Qualifying, Race, Test), car, driver, date/time, weather (dry/wet), air and track temperature, target lap count, run plan notes.
Process	INSERT Session linked to Event, Car, and Driver; allow edits until the engineer locks the session.
Output	Session page showing its setup, tires, laps, and driver feedback.

FR: 7 Setup Sheet Recording

Description	Store the exact car setup used in a session.
Input	Front/rear wing angle, ride height, tire pressures (4 corners), camber, toe, spring and damper settings, anti-roll bar, gearing, fuel load, notes. Option to clone a previous setup.
Process	INSERT SetupSheet linked to the Session; when cloning, SELECT the old setup and copy its values into the new row.
Output	Saved setup sheet shown on the session page.

FR: 8 Tire Set Usage and Heat Cycles

Description	Record which tire set was used and track tire life.
Input	Tire set, session, laps run on the set.
Process	INSERT SessionTireUse; UPDATE TireSet to increase its heat-cycle count by one; if the count reaches the set's maximum, mark it Retired; warn when it is within one cycle of the maximum.
Output	Updated heat-cycle count, remaining life, and a warning banner if needed.

FR: 9 Lap Time Entry

Description	Add lap times for a session.
Input	Lap number and lap time (optional sector times) entered manually, or a CSV file exported from a timing system.
Process	Validate format, reject impossible values, INSERT Lap rows in one transaction; calculate best lap and average lap for the session.
Output	Lap table and lap-time chart; import summary (rows accepted/rejected).

FR: 10 Best-Setup Search

Description	Find proven baseline setups for an upcoming event.
Input	Track, weather, track temperature range, car class.
Process	JOIN Session, Event, SetupSheet, Car, and Lap; filter by the inputs and rank sessions by best lap time.
Output	Ranked list of setups; the engineer can clone one into a new session (FR-7).

3.5 Driver

FR: 11 View Run Plan and Submit Feedback

Description	Drivers see what to do in their session and report how the car felt afterward.
Input	Selected session; feedback text (e.g., "understeer in Turn 3 on entry").
Process	SELECT sessions assigned to the logged-in driver with run plan notes, tire set, and target lap count; UPDATE the Session's driver feedback field.
Output	Read-only run plan page (mobile friendly); saved feedback shown to the engineer on the session page.

FR: 12 Personal Lap Analysis

Description	Drivers track their own progress.
Input	Track and date range.
Process	SELECT the driver's laps by joining Lap, Session, and Event; compute best lap, average, and trend by event.
Output	Personal best table and progress chart.

3.6 System Administrator

FR: 13 Team Approval and Track Catalog

Description	Control which teams can use the platform and maintain the shared track list.
Input	Pending team and action (approve, suspend, reactivate); track name, location, length, number of turns.
Process	UPDATE Team status (suspended teams cannot log in); INSERT/UPDATE Track; prevent deleting a track already used by an event.
Output	Team list with status; updated track catalog available in all teams' drop-down lists.

4.3 Access Control

Pitwall uses **role-based access control (RBAC)**. A servlet filter checks the user's role before every request, and every query includes the user's team ID so a team can never read or change another team's data. Records are soft-deleted (deactivated) to keep history.

Function	Driver	Engineer	Manager	Admin
Login / own profile (FR-1, FR-2)	Yes	Yes	Yes	Yes
Roster, cars, tire sets, events (FR-3 to FR-5)	View only	View only	Full	No
Sessions, setups, tire usage, laps (FR-6 to FR-9)	View own	Full	Full	No
Best-setup search (FR-10)	No	Full	Full	No
Run plan and feedback (FR-11)	Full (own)	View all	View all	No
Personal lap analysis (FR-12)	Own	All drivers	All drivers	No
Team approval and track catalog (FR-13)	No	No	No	Full